

METHOD OF MIXTURES (6-A)

Aim

Determination of the specific heat capacity of brass by the method of mixtures

Materials and Apparatus

Tripod with mesh, water beaker, brass sphere, two thermometers, stands, Bunsen burner, wooden pad, two calorimeters one with insulation, stirrer, wooden lid, stop watch and electronic balance

Theory

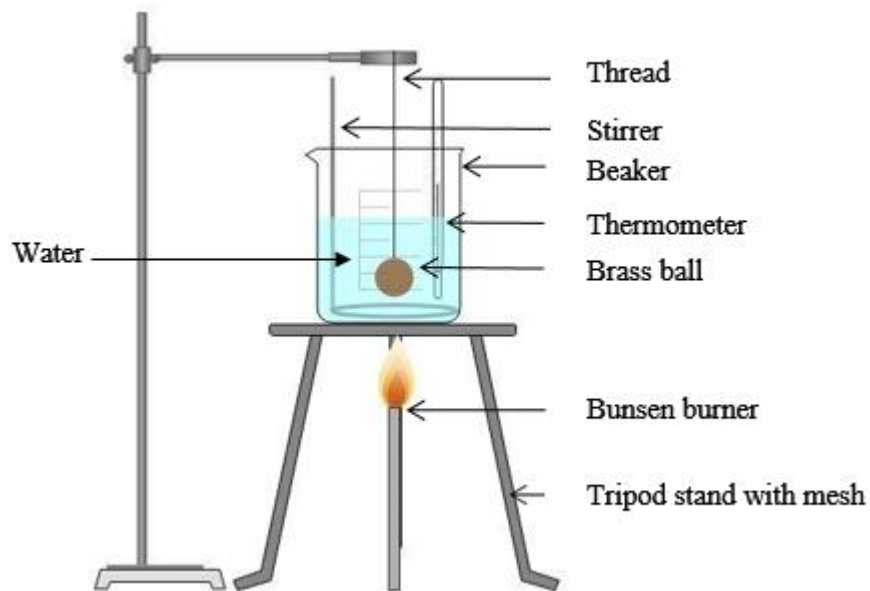


Fig. 1: Experimental Setup of heating the brass ball

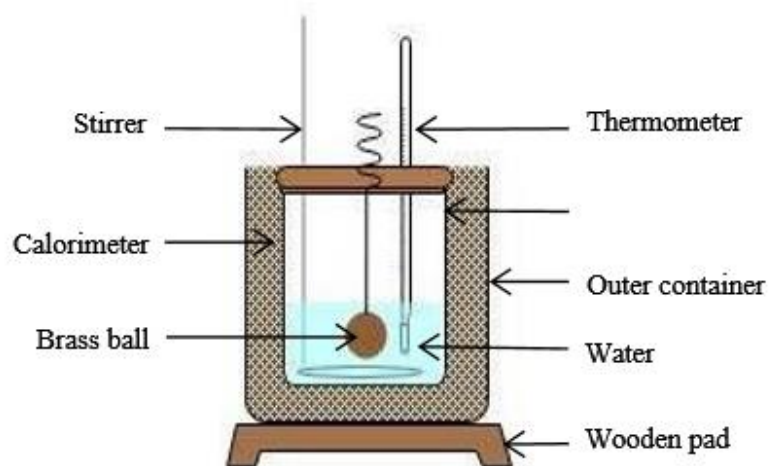


Fig. 2: Experimental setup to determine the specific heat capacity of brass

Setup Heat gained or lost by an object can be written as:

$$\Delta Q = m C \Delta \theta \quad (1)$$

where,

m : Mass of the object

C : Specific heat capacity of the material

$\Delta \theta$: Temperature difference

When the heated brass ball is transferred to the calorimeter with water.

Heat lost by the brass ball = Heat gained by the calorimeter + water inside

$$M C (\theta_1 - \theta_2) = (M_c C_c + M_w C) (\theta_2 - \theta_0) \quad (2)$$

$$C = (M_c C_c + M_w C_w) (\theta_2 - \theta_0) / [M (\theta_1 - \theta_2)] \quad (3)$$

where,

C : Specific heat capacity of brass

C_c : Specific heat capacity of calorimeter material

C_w : Specific heat capacity of water

M_c : Mass of the calorimeter with stirrer

M : Mass of brass ball

M_w : Mass of water

θ_1 : Initial temperature of the brass ball

θ_2 : Final temperature of the brass ball with water

θ_0 : Initial temperature of water

Procedure

a) Measure the mass of the brass ball and the mass of the calorimeter with the stirrer.

Q: What is the advantage of having a calorimeter and a stirrer made of the same material?

b) Fill the calorimeter with water.

Q: How much of the calorimeter's height should be filled with water?

c) Measure again the mass of the calorimeter with water.

d) Hang the brass ball and immerse it in a water beaker (as in Fig. 1).

e) Heat the water up to its boiling point.

Q: What is the precaution that should be taken when hanging the ball and why?

Q: How do you ensure uniform heating?

f) Take down the temperature of boiling water when the steady state is reached.

g) Transfer the ball into the calorimeter with water (as in Fig. 2).

Q: Why is it important to transfer the brass ball **quickly** into the calorimeter?

h) While stirring, note down the temperature of the mixture in every 15 seconds over a period of about 15 minutes.

METHOD OF MIXTURE (6-B)

Aim

Determination of the heat capacity of an unknown liquid

Apparatus

Tripod with mesh, water beaker, brass sphere, two thermometers, stands, Bunsen burner, wooden pad, two calorimeters one with insulation, stirrer, wooden lid, unknown liquid, stop watch, and electronic balance

Theory

The heat lost by the brass equals the heat gained by the unknown liquid:

Using the equation (1);

$$M \times C(\theta_1 - \theta_2) = (M_c C_c + M_l C) (\theta_2 - \theta_0) \quad (4)$$

where,

C : Specific heat capacity of brass

C_c : Specific heat capacity of calorimeter material

C_l : Specific heat capacity of the unknown liquid

M_c : Mass of the calorimeter with stirrer

M : Mass of brass ball

M_l : Mass of the unknown liquid

θ_1 : Initial temperature of the brass ball

θ_2 : Final temperature of the brass ball with the unknown liquid

θ_0 : Initial temperature of the unknown liquid

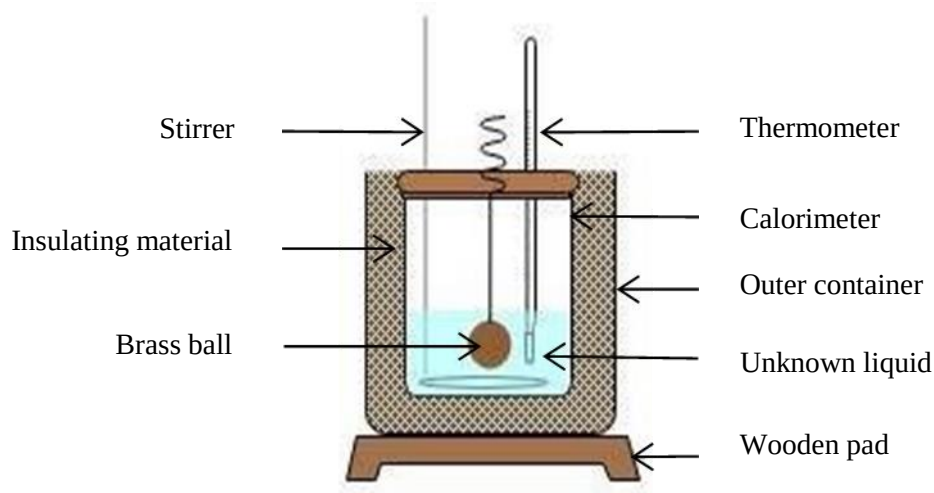


Fig. 3: Experimental setup for determination of specific heat capacity of an unknown liquid

METHOD OF MIXTURE (6-B)**Procedure**

- a) Measure the mass of the brass ball and the mass of the calorimeter with the stirrer.
- b) Fill the calorimeter with the given liquid.
- c) Measure again the mass of the calorimeter with the liquid.
- d) Hang the brass ball and immerse it in a water beaker (as in Fig. 1).
- e) Heat the water up to its boiling point.
- f) Take down the temperature of boiling water when the steady state is reached.
- g) Transfer the ball into the calorimeter with the liquid (as in Fig. 3).
- h) While stirring, note down the temperature of the mixture in every 15 seconds over a period of about 15 minutes.